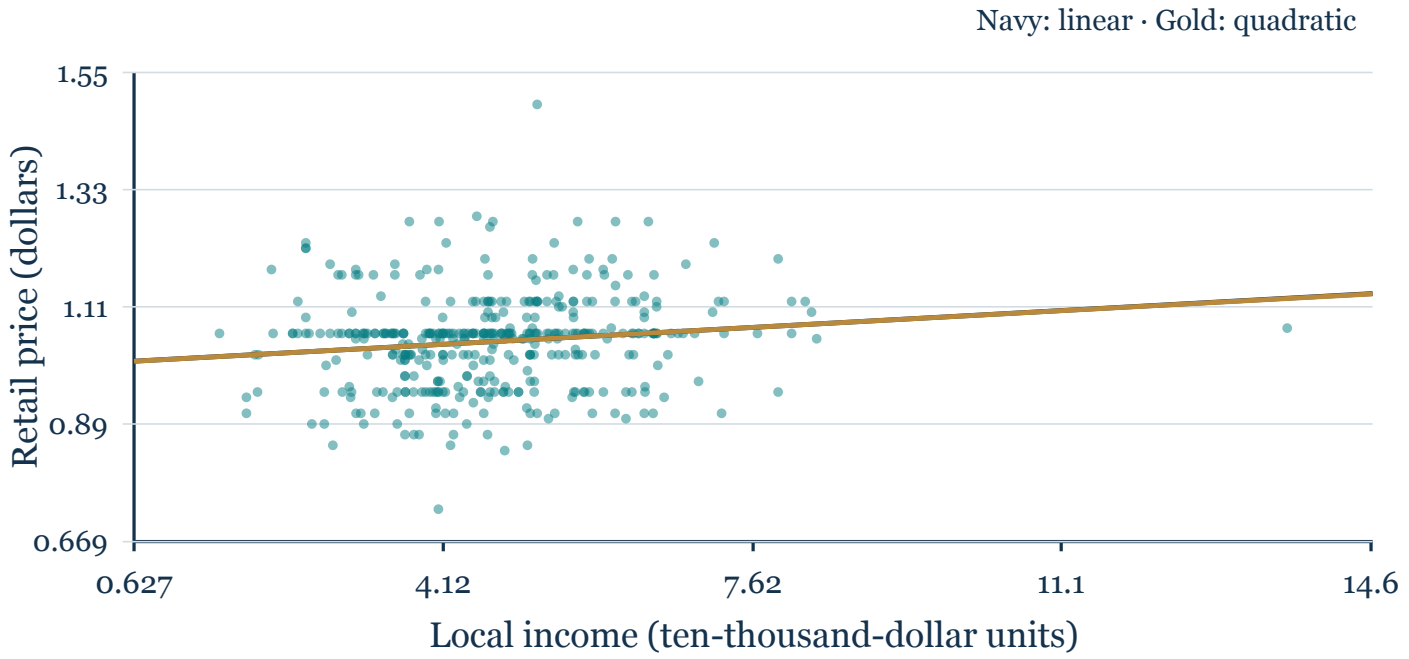
Do prices differ by the same amount or the same percent?

Choose a scale that answers the economic question.



401 observed positive-price and positive-income records. Lines use the actual level and quadratic regressions; no geographic sampling claim is made.

Define the observation and the usable sample

Inspect the data before interpreting a transformed variable.

> $X = \text{income}/10{,}000$

> Outcome: medium soda price, in dollars

A level–level model has a constant unit slope

A one-unit income change produces the same fitted dollar difference.

```
. regress sprice inc10, vce(robust)

------------------------------------------------------------------------------
             |               Robust
      sprice | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       inc10 |   .0090733   .0031523     2.88   0.004     .0028761    .0152705
       _cons |   1.002219   .0157154    63.77   0.000     .9713236    1.033114
------------------------------------------------------------------------------
```

$$\widehat{P} = a + bX$$

$$\Delta\widehat{P} = b\Delta X$$

A logarithm expresses a proportional scale

Small log differences approximate percentage differences.

> $\log(1) = 0, \quad \log(2) \approx 0.693, \quad \exp(0.693) \approx 2$

> $\log(2.20) - \log(2) = \log(1.10) \approx 0.09531$

> $100\{\exp(0.09531) - 1\} \approx 10\%$

A log–level model gives a proportional outcome contrast

Keep the regressor units visible.

> $\widehat{\log P} = a + bX$

> $100\{\exp(b\Delta X) - 1\}$

Which average does a log fit describe?

An average log is not the log of an arithmetic average.

> Arithmetic mean: $(1 + 9)/2 = 5$

> Exponentiated mean log: $\exp\{(\log 1 + \log 9)/2\} = 3$

> State which prediction quantity is being compared.

Exact log change

Try the comparison before opening the explanation.

Practice

In a log-outcome model, a policy dummy has coefficient 0.08. Give the usual approximate percentage contrast and the exact multiplicative contrast in the fitted exp(linear predictor).

Exact log change: explanation

Follow the calculation and interpret its units.

> Approximate: +8%. Exact: $100[\exp(0.08) - 1] = 8.3287\%$. This is a contrast in the exponentiated fitted log outcome. A conditional-mean claim in outcome levels additionally needs a justified retransformation treatment of the error distribution.

A level–log model gives a level outcome contrast

A percent change in X becomes a small change in log X.

$$\widehat{Y} = a + b \log X$$

$$\Delta\widehat{Y} = b \log(X_1/X_0)$$

A log–log slope is an elasticity

It relates proportional changes in both variables.

> $\widehat{\log Y} = a + b \log X$

> $\%\Delta \widehat{Y} = 100\{(X_1/X_0)^b - 1\}$

Elasticity

Try the comparison before opening the explanation.

Practice

In $\log(quantity) = a - 0.7\log(price) + u$, what is the local elasticity? What is the exact fitted quantity change for a 10% price rise?

Elasticity: explanation

Follow the calculation and interpret its units.

> The elasticity is −0.7. A 10% price rise multiplies fitted exponentiated log quantity by $1.1^{-0.7}$; the percentage contrast is $100[1.1^{-0.7} - 1]$, about −6.45%, not exactly −7%. Causal interpretation requires price exogeneity or a valid identification strategy.

Read the actual price–income elasticity

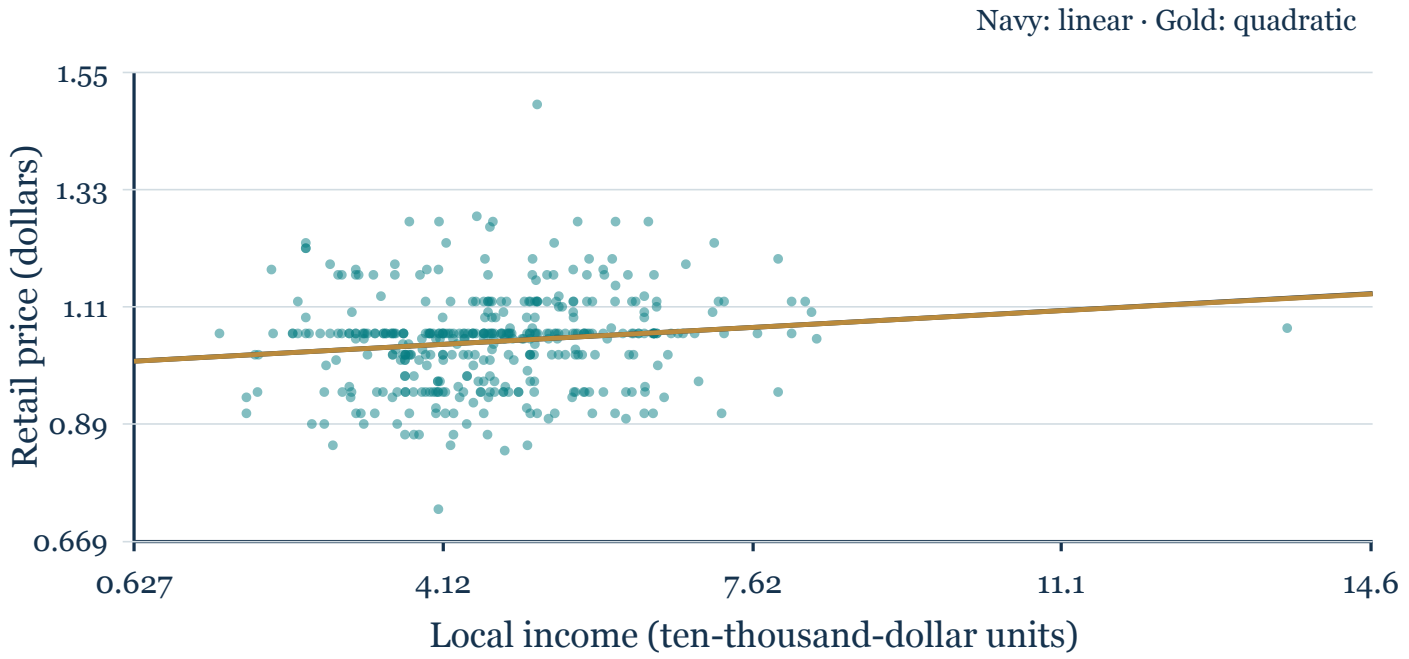
A unit-free slope still needs an empirical interpretation.

```
. regress lp li, vce(robust)

------------------------------------------------------------------------------
             |               Robust
          lp | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
          li |   .0374749   .0145083     2.58   0.010     .0089527    .0659971
       _cons |  -.3613775   .1559216    -2.32   0.021     -.667908   -.0548471
------------------------------------------------------------------------------
```

Compare predictions for the same realistic change

Use the equation, not just the coefficient label.



401 observed positive-price and positive-income records. Lines use the actual level and quadratic regressions; no geographic sampling claim is made.

A quadratic lets the slope change with X

Define the squared term before interpreting it.

> $\widehat{Y} = 500 + 8X - 0.2X^2$

> Local slope: $d\widehat{Y}/dX = 8 - 0.4X$

> General slope: $b_1 + 2b_2X$

Use a finite quadratic difference for a finite change

The tangent slope is a local description.

> $\widehat{Y}(10) = 560, \quad \widehat{Y}(15) = 575, \quad \Delta\widehat{Y} = 15$

> $\Delta\widehat{Y} = b_1\Delta + b_2(2X\Delta + \Delta^2)$

> Local approximation: $4 \times 5 = 20$, not the finite difference.

A turning point must be inside a meaningful range

Do not report an unsupported optimum.

> $X^* = -\dfrac{b_1}{2b_2}$ when $b_2 \neq 0$

Quadratic slope

Try the comparison before opening the explanation.

Practice

For $\widehat{score} = 500 + 8X - 0.2X^2$, give the slope at $X = 10$ and the turning point. What must be checked before using the turning point?

Quadratic slope: explanation

Follow the calculation and interpret its units.

> The slope is $8 - 0.4X$, which equals 4 at $X = 10$. The turning point is $X = 20$, a maximum because the squared coefficient is negative. Check that $X = 20$ lies within the observed and substantively meaningful range; extrapolated turning points are unreliable.

Inspect the actual curvature rather than assume it

A flexible model may provide little evidence of bending.

```
. regress sprice c.inc10##c.inc10, vce(robust)

------------------------------------------------------------------------------
             |               Robust
      sprice | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       inc10 |   .0091573   .0103187     0.89   0.375    -.0111286    .0294432
             |
c.inc10#c.inc10 |  -7.86e-06   .0008332    -0.01   0.992    -.0016458    .0016301
             |
       _cons |   1.002011   .0304353    32.92   0.000     .9421772    1.061845
------------------------------------------------------------------------------
```

Ask Stata for slopes at stated values

margins makes the comparison explicit.

```
. margins, dydx(inc10) at(inc10=(2 4 6))

Conditional marginal effects                                    Number of obs = 401
Model VCE: Robust

Expression: Linear prediction, predict()
dy/dx wrt:  inc10
1._at: inc10 = 2
2._at: inc10 = 4
3._at: inc10 = 6

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
inc10        |
         _at |
          1  |   .0091259   .0071961     1.27   0.205    -.0050213    .0232731
          2  |   .0090945    .004394     2.07   0.039      .000456    .0177329
          3  |    .009063   .0030074     3.01   0.003     .0031507    .0149754
------------------------------------------------------------------------------
```

Retransforming a log fit needs care

The exponential of an average log is not the average level.

> $\log Y = m(X) + u$

> $E(Y \mid X) = \exp\{m(X)\}\, E\{\exp(u) \mid X\}$

> The retransformation factor need not be one or constant across X.

Zeros and negative values are substantive information

Do not hide them through an automatic log transformation.

> The natural log is defined only for positive values.

> In this retail application, the analysis explicitly requires positive observed price and income.

Choose a form for a decision-sized comparison

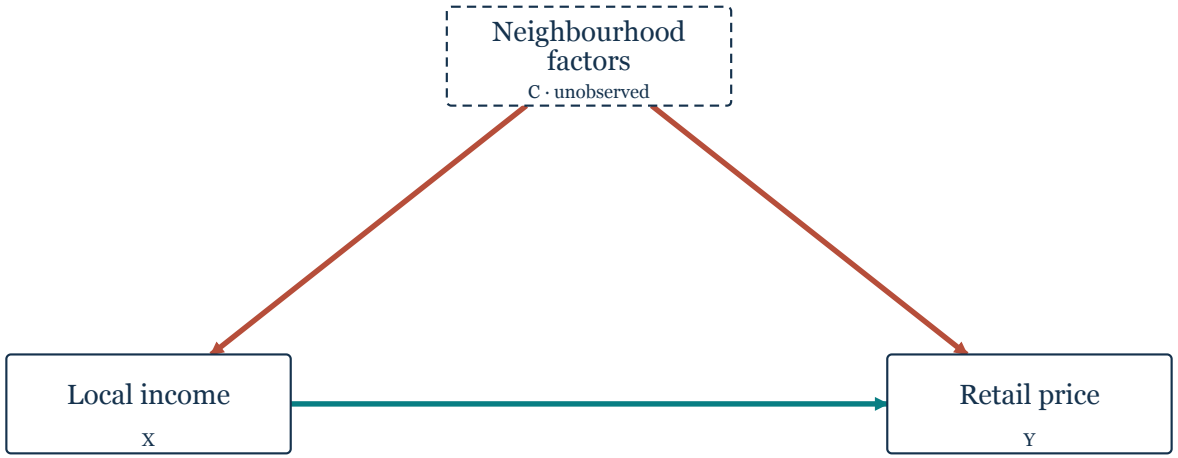
Keep fit, interpretation and design connected.

> An analyst studying affordability may want dollar price differences.

> Use graphs and stated prediction contrasts to communicate the model.

Transforming the variables does not close a backdoor

A better scale and a credible comparison answer different questions.



1 open directed route · 1 open alternative · gold: conditioned on

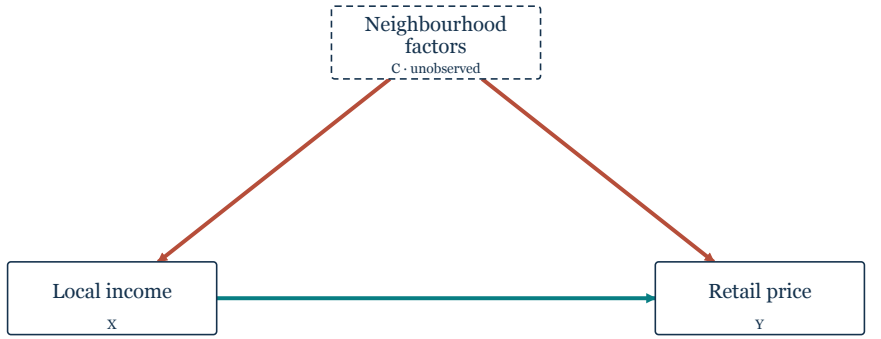
Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

> Functional form chooses the scale and shape.

> Identification addresses the alternative causal explanations.

DAG practice: logs and a common cause

Trace the routes before choosing a control.



1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

Neighbourhood characteristics influence local income and retail prices. Would replacing prices and income by their logarithms close the common-cause route? Explain the distinct roles of the graph and the functional form.

DAG practice: logs and a common cause: explanation

Explain what conditioning changes in this stated graph.

> The route $X \leftarrow C \rightarrow Y$ remains. A numerical transformation does not remove the economic mechanism represented by C. A better fit or a convenient elasticity therefore does not establish causal income variation. The graph states the proposed causal relationships and helps assess controls or alternative designs. The equation chooses how a given comparison is represented numerically and which finite or local contrast is reported. An applied analysis needs both a defensible comparison and a useful specification.

Translate retail coefficients into a useful contrast

Reproduce the evidence, then explain its economic meaning.

Practice

Change the income comparison from 30,000 to 50,000 dollars. Report the level-model dollar contrast and the exact exponentiated-log percentage contrast. Keep your commands, log, output and interpretation.

Translate retail coefficients into a useful contrast: explanation

A reproducible result needs both a calculation and an interpretation.

> A 20,000-dollar income difference gives a 0.018147-dollar level-model contrast.

> From 30,000 to 50,000: exponentiated-log contrast 1.932756%.

> A fitted log-value ratio needs extra assumptions to be an arithmetic-mean ratio.

Connect functional form with the next data-quality question

A better curve cannot repair every data problem.

> We can now translate coefficients in levels, logs and quadratic models into meaningful changes.

> The next lesson asks what happens when we collect more observations while the design or measurements remain imperfect.